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Institute of Management & Career Courses (IMCC)

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DEPARTMENT OF COMPUTER APPLICATIONS (MCA)

CAPSTONE PROJECT SCOPE PROPOSAL & TECHNICAL SYNOPSIS

PENDING FACULTY REVIEW

Academic Cohort: **2026-2028** • Class: **FY MCA** • Section: **Division A** • **Group 27**

Project Title	**"InterviewIQ" - An Explainable AI Mock Interview and Placement Coach - Practice like it is real. Improve with evidence.**		
Team Name	**Quantum Minds** `Group 27`	**Project Track / Domain**	Artificial Intelligence & Machine Learning
Project Category	Capstone Team Project	**Submission Date**	8 October 2026
Internal Faculty Guide	**Dr. Shweta Meshram** sdm.imcc@mespune.in	**External Industry Guide**	**AUTO** Industry Organization

1. STUDENT PROJECT TEAM ROSTER

TOTAL MEMBERS: 2

#	STUDENT FULL NAME	ROLL NO. / PRN	INSTITUTIONAL EMAIL	PROJECT RESPONSIBILITY
1	**Manas C. Deshmukh**	2601042	manas.deshmukh.work@gmail.com	Lead **Team Leader (Lead)**
2	**Swayam G. Kadam**	2601088	kadamswayam29@gmail.com	**Core Team Member**

2. PROBLEM STATEMENT & ACADEMIC SYNOPSIS

Candidates preparing for job interviews have little access to realistic, honest practice. Friends are too polite, mentors have limited time, and generic question lists give no feedback, so many candidates freeze or ramble under pressure, struggle to explain their own projects, and do not know what is holding them back.

Existing AI mock-interview tools offer voice interviews and filler-word feedback, but they rarely explain how an answer was scored or show that their scores agree with human evaluators, so candidates receive scores they cannot trust or act on.

There is a need for a practice platform that acts like a real interviewer, asks questions based on the candidate's role and resume, and gives explainable feedback on both what they say and how they say it, with tracked progress over time.

3. PROJECT SCOPE & FUNCTIONAL OBJECTIVES

To build a web-based AI interview practice platform that acts like a real interviewer, asks questions based on the candidate's role and resume, and gives explainable feedback on what they say and how they say it, so any candidate can see what to improve and track their progress.

Key objectives :-
Provide realistic voice-based interviews with follow-up questions, in the candidate's chosen category, from the built-in or their own question bank.
Score answers with an explainable rubric (correctness, depth, clarity, examples), validated against human ratings.
Analyse delivery (response delay, pace, filler words, pauses) against the candidate's own baseline.
Keep a complete interview history, with category-wise progress reports and recordings the candidate can replay and download.
Protect privacy through consent, opt-in recording and the ability to delete data.

4. TECHNOLOGY STACK & INFRASTRUCTURE ENGINES

PRIMARY DEVELOPMENT & ARCHITECTURAL STACK

HTML5 CSS3 JavaScript Python FastAPI WebSocket

DATABASES, CACHES & STORAGE ENGINES

SQLite MySQL

5. ARCHITECTURAL MODULE BREAKDOWN & DELIVERABLES

8 DEFINED MODULES

#	FUNCTIONAL MODULE	DETAILED SYSTEM SPECIFICATION & DELIVERABLE SCOPE
M1	**Module 1: User Management and Interview Setup**	Sign-up and login with hashed passwords, email verification, role-based access (Candidate, Mentor, Administrator) and consent. The candidate chooses role, level, interview category (such as Python, SQL, aptitude, behavioural or HR), type, persona, mode (practice, Project Defense, Interview Drive) and an optional timer. They pick the built-in or their own question bank, assign the interview to a project, choose whether to save a recording, and can upload a text-based PDF or DOCX resume. The resume parser extracts skills and projects to generate personalised questions.
M2	**Module 2: Question Bank and Role-Based Retrieval**	Questions are stored by role and category with topic, difficulty, reference answer and key concepts, each with a sentence embedding. The system retrieves the closest questions and reference answers for the chosen role, and K-Means groups questions into topics to find weak areas. Candidates can add their own question bank (CSV, Excel or form), and every custom question must include a reference answer. Custom banks are private to the candidate.
M3	**Module 3: Real-Time AI Interviewer**	Runs the live voice or text interview over a WebSocket, controlled by a state machine. It speaks like a human interviewer, with short replies, acknowledgments, interruption handling and silence prompts. After each answer it cross-questions: it goes deeper, probes, challenges, hints or redirects, and flags contradictions, including answers that don't match the resume. This module also covers Project Defense mode, the multi-round Interview Drive simulator, language and accessibility support (Indian English and Hinglish, a stammer-friendly mode, a low-bandwidth fallback), and optional recording if the candidate has opted in.
M4	**Module 4: Answer Analysis and Scoring**	Compares each answer with the retrieved reference answer and key concepts, and scores correctness, depth, clarity and use of examples. It lists covered, missing and incorrect concepts and checks STAR structure on behavioural questions. An ML score predictor (Random Forest or Gradient Boosting) is trained on human-rated answers and validated by correlation and error. Scores on custom questions are labelled as based on the candidate's own reference answer.
M5	**Module 5: Speech and Delivery Analytics**	Measures response delay, answer time, speaking pace, filler words, pauses, hedging and restarts for each question, compared with the candidate's own baseline. Each question is placed as correct or wrong, steady or shaky. These are delivery indicators, not measures of emotion.
M6	**Module 6: Skill Tracking, Progress and Reports**	Saves every session in MySQL, with recordings stored as files and only their paths in the database. Candidates get a complete interview history with filters and reopenable reports. The system tracks skill per topic, adjusts difficulty, repeats weak questions with spaced repetition, and gives a readiness score and a progress forecast. A category-wise progress report shows interviews, scores and improvement per category. Interviews can be grouped into projects and compared. Reports can be downloaded as PDF or emailed, and with consent the recording can be replayed by question and downloaded.
M7	**Module 7: Mentor Review and Sharing**	The candidate shares a report through a secure link, and a mentor reads it and adds comments on individual questions.
M8	**Module 8: Admin, Validation and Privacy**	The administrator manages users, questions and rubrics, and enters human ratings for model training. A validation page shows agreement with human raters, follow-up accuracy, response time and known limitations, along with an anonymised weakest-topics chart. Privacy controls include explicit consent before recording, opt-in recordings private to the candidate, a retention period, data export, and deletion of an interview together with its recording files.

6. TANGIBLE DELIVERABLES & EXPECTED OUTPUT

A working web application that conducts voice interviews based on the candidate's role and resume, and gives each candidate:

Explainable question-wise feedback: scores for correctness, depth, clarity and examples, with covered and missed concepts and a model answer.
Delivery analysis: response delay, pace, filler words and pauses, compared with their own baseline.
A trusted score: an ML-predicted score validated against human ratings.
Progress tracking: skill per topic, category-wise reports, complete interview history by project, and a readiness forecast.
Optional recording: replay and download for self-review.
Reports that can be downloaded as a PDF, emailed, or shared with a mentor for comments.

FACULTY SUPERVISOR REVIEW & AUDIT ASSESSMENT:

add questions bank, recording sent to user,

Manas C. Deshmukh	**Dr. Shweta Meshram**	**AUTO**	**Head of Department**
Lead Sign-off / Date	Internal Faculty Supervisor	External Industry Mentor	Dept. of Computer Applications (MCA)

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